V ⊗ W
V ⊕ W
A ↓ B
∏
∑_n

Components

X	object
f: X → Y	arrow
A	category
F: A → B ∈ B^A	functor i.e. a path
B^A = { A → B }	functor category
A(X, Y) ∈ Set	hom set
A(-, Y): A → Set ∈ Set^A	hom functor
φ: F ⇒ G	natural transformation
Nat(F, G) = { F ⇒ G }	

==========

Adjoint

[Stackexchange] How to recognize adjoints: https://math.stackexchange.com/questions/742640/how-to-recognize-adjointness
[Stackexchange] why should I care about adjoint functors: https://math.stackexchange.com/questions/25455/why-should-i-care-about-adjoint-functors
[Mathoverflow] what is an intuitive view of adjoints: https://mathoverflow.net/questions/6551/what-is-an-intuitive-view-of-adjoints-version-1-category-theory

Interpretation 1: weak isomorphism
An adjoint is a **weak isomorphic functor**

	Isomorphic functor G: B → A	**Adjoint φ: B → A**
Arrows	G: { -GX → Y } → { X → GY }	φ: { FX → Y } → { X → GY }
	Gh: X → GY	φh: X → GY
Composition		
f: X' → X ∈ A	G(h ∘ -Gf) = Gh ∘ f: X' → GY	φ(h ∘ Ff) = φh ∘ f: X' → GY
g: Y → Y ' ∈ B	G(g ∘ h) = Gh ∘ Gh: X' → GZ	φ(g ∘ h) = Gg ∘ φh: X → GY'

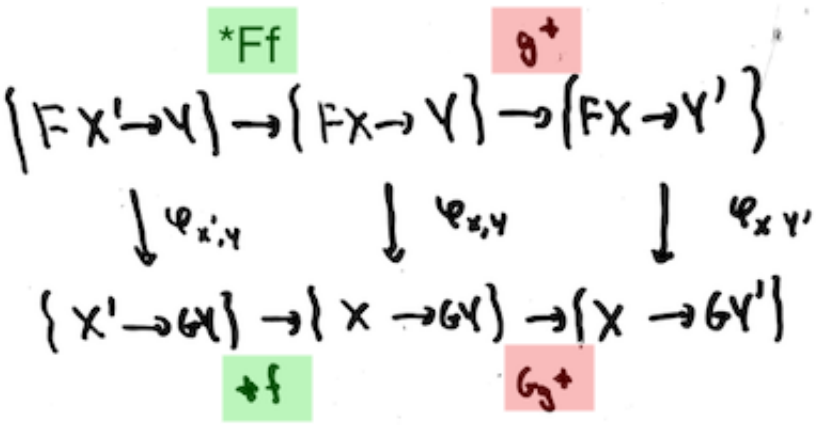
Interpretation 2: universal property natural in both variables
An adjoint is a **universal property natural** in both X and Y

Adjoint
φ: B → A

Universal property
$\varphi X,Y$: { FX → Y } → { X → GY }

natural in X
$\varphi X',Y \circ {}^*Ff$ = $f^* \circ \varphi X,Y$
$\varphi X',Y(h \circ Ff)$ = $\varphi X,Y(h) \circ f$

natural in Y
$\varphi X,Y' \circ g^*$ = $Gg^* \circ \varphi X,Y$
$\varphi X,Y'(g \circ h)$ = $Gg \circ \varphi X,Y(h)$



Interpretation 3: biuniversal property
An adjoint is a **biuniversal property** i.e. universal in both directions

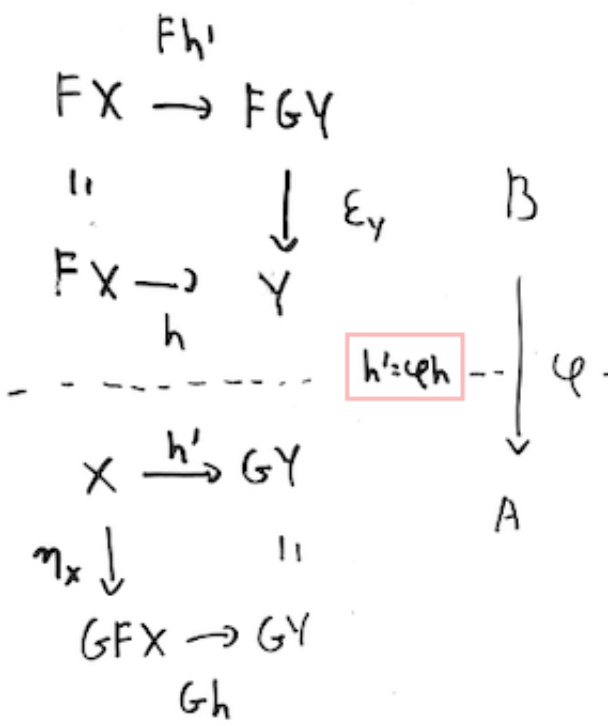
Biunivesal property

	B		A
φ:	{ h: FX → Y }	<=>	{ h': X → GY }

Gh $\circ$ η**X = h'**
h = ε**Y** $\circ$ **Fh'**

Universal into G
ηX: X → GFX **universal arrow**
φh: X → GY
φh = Gh $\circ$ ηX

Couniversal into F
ε: FGY → Y co**universal arrow**
-φh': FX → Y
-φh' = εY $\circ$ Fh'



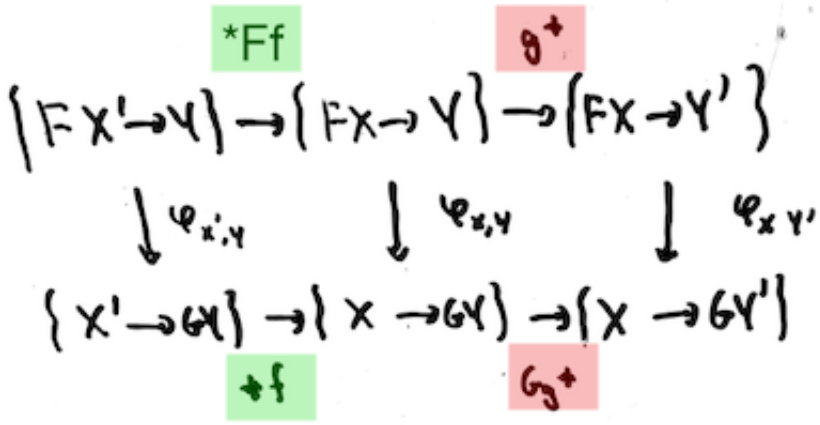
Definition
Adjoint
Let functors F: A → B, G: B → A
If there exists

$$\varphi X,Y: \{ FX \xrightarrow{B} Y \} \Rightarrow \{ X \xrightarrow{A} GY \}$$

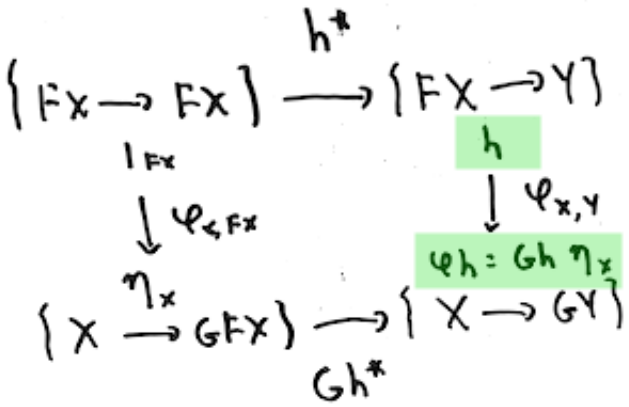
where φ is:

a) **bijective**
b) **natural** in X and Y (as homsets)
$\varphi X',Y \circ {}^*Ff = {}^*f \circ \varphi X,Y \qquad f: X \to X'$
$\varphi X,Y' \circ g^* = Gg^* \circ \varphi X,Y \qquad g: Y \to Y'$

Then **<F, G, φ>: A → B** is an **adjunction from A to B.**



Universality φ: {FX→Y} → {X→GY}
$\varphi h = Gh \circ \eta X$



Terminology

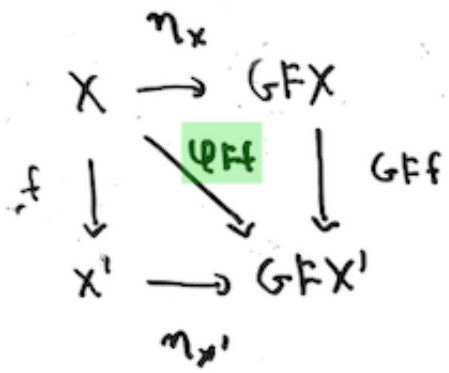
B A
φ: { FX → Y } ⇒ { X → GY }

F: A → B	**Left adjoint** of G
G: B → A	**Right adjoint** of F
ηX: X → GFX ∈ A = φ1FX	**unit**
εY: FGY → Y ∈ B = φ^-1(1GY)	**counit**
<F, G, φ>: A → B where φ: { FX → Y } ⇒ { X → GY } B A	**Adjunction** with transformation φ
<F, G, η , ε>: A → B where unit η**: IA ⇒ GF** counit ε**: FG ⇒ IB**	**Adjunction** induced by **units** η and ε

Properties
- **universal/couniversal property**

natural unit aka universal arrow
η: IA ⇒ GF
ηX: X→GFX

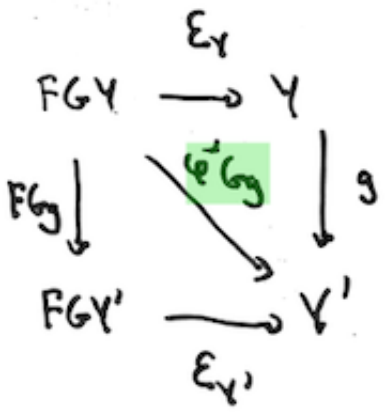
ηX	= φ1FX	definition
ηX ∘ f	= GFf ∘ ηX'	naturality f:X → X'
	= φFf	



natural counit aka couniversal arrow
ε: FG ⇒ IB

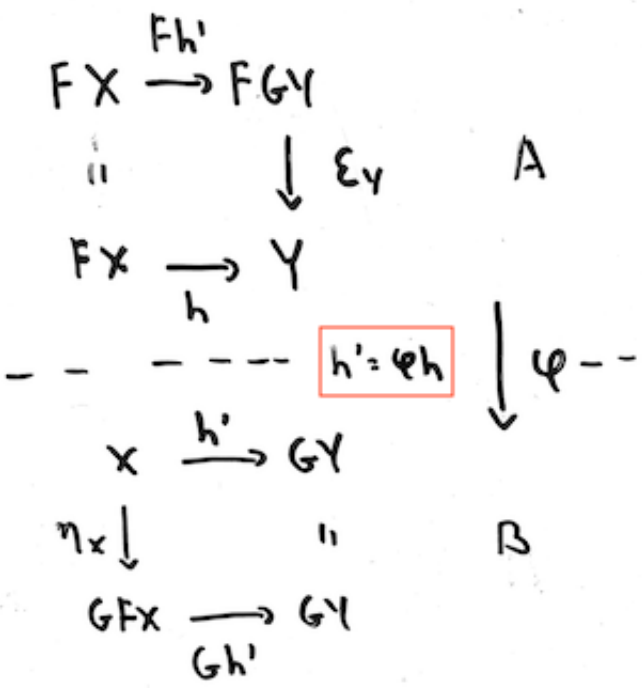
εY: FGY → Y

εY	=	$-\varphi 1GY$	definition
$g \circ \varepsilon Y$	=	$\varepsilon Y' \circ FGg$	naturality g:Y → Y'
	=	$-\varphi Gg$	

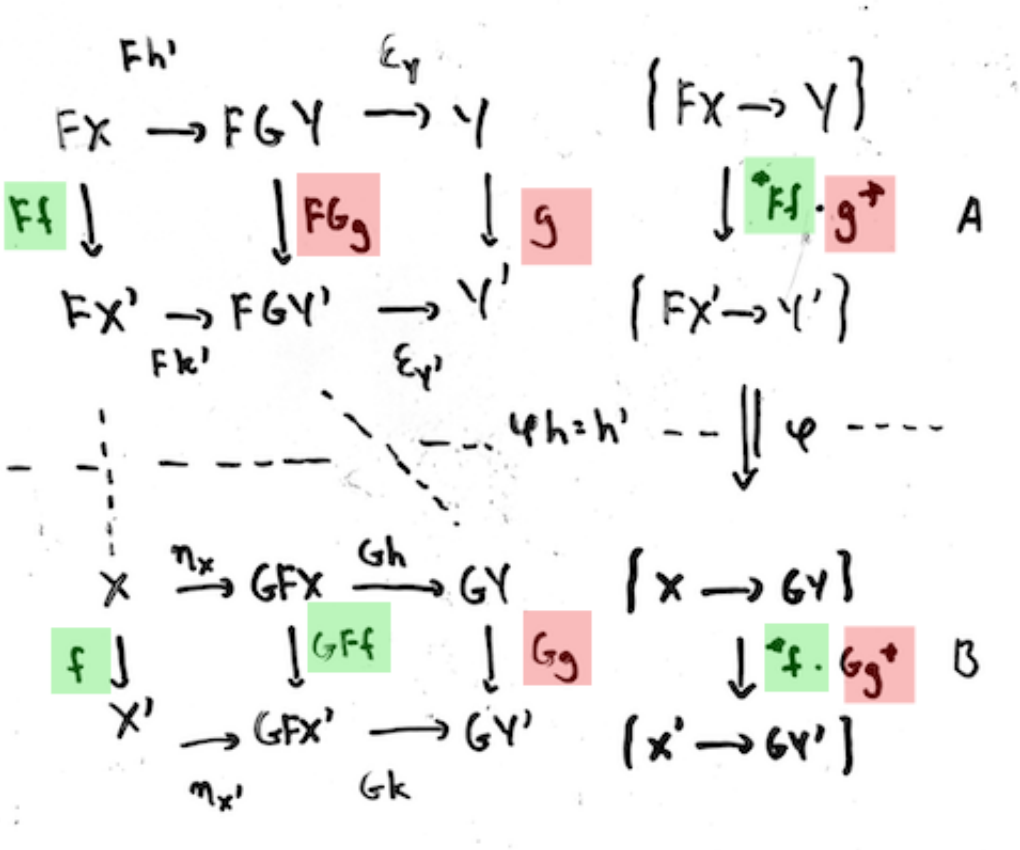


universal properties

φh: X→GY	university property of G
φh = Gh ∘ ηX	h:FX → Y
$-\varphi h'$: FX→Y	university property of F
$-\varphi h'$ = $\varepsilon Y \circ Fh'$	h': X → GY



naturality



Identities
h: FX → Y

h	=	$-\varphi(\varphi h)$	
	=	$-\varphi(Gh \circ \eta X)$	universality of φ
	=	$\varepsilon Y \circ F(Gh \circ \eta X)$	universality of $-\varphi$
	=	$\varepsilon \mathbf{Y} \circ \mathbf{FGh} \circ \mathbf{F}\eta\mathbf{X}$	
	=	$(h \circ \varepsilon \mathbf{FX}) \circ \mathbf{F}\eta\mathbf{X}$	by naturality of ε
	=	h	
		i.e. preimage of h	
1FX	=	$\varepsilon \mathbf{FX} \circ \mathbf{F}\eta\mathbf{X}$	with h = 1FX

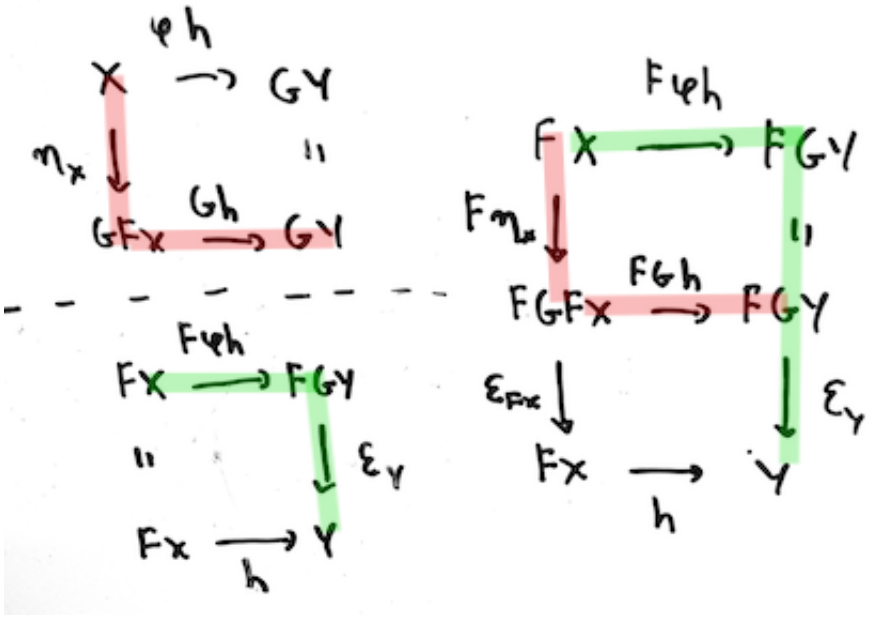
h' = φh

h'	=	$\varphi(-\varphi(h'))$	
	=	$\varphi(\varepsilon Y \circ Fh')$	universality of φ
	=	$G(\varepsilon Y \circ Fh') \circ \eta X$	universality of $-\varphi$
	=	$\mathbf{G}\varepsilon\mathbf{Y} \circ \mathbf{GF}h' \circ \eta\mathbf{X}$	
	=	$G\varepsilon Y \circ (\eta GY \circ h')$	by naturality of η
	=	h'	
1GY	=	$\mathbf{G}\varepsilon\mathbf{Y} \circ \eta\mathbf{GY}$	with h'=1GY

Preimages

h	=	$\mathbf{-\varphi(Gh \circ \eta X)}$	
	=	$h \circ -\varphi(\eta X)$	by naturality
	=	h	
h'	=	$\varphi(\varepsilon \mathbf{Y} \circ \mathbf{Fh'})$	
	=	$\varphi(\varepsilon Y) \circ h'$	by naturality

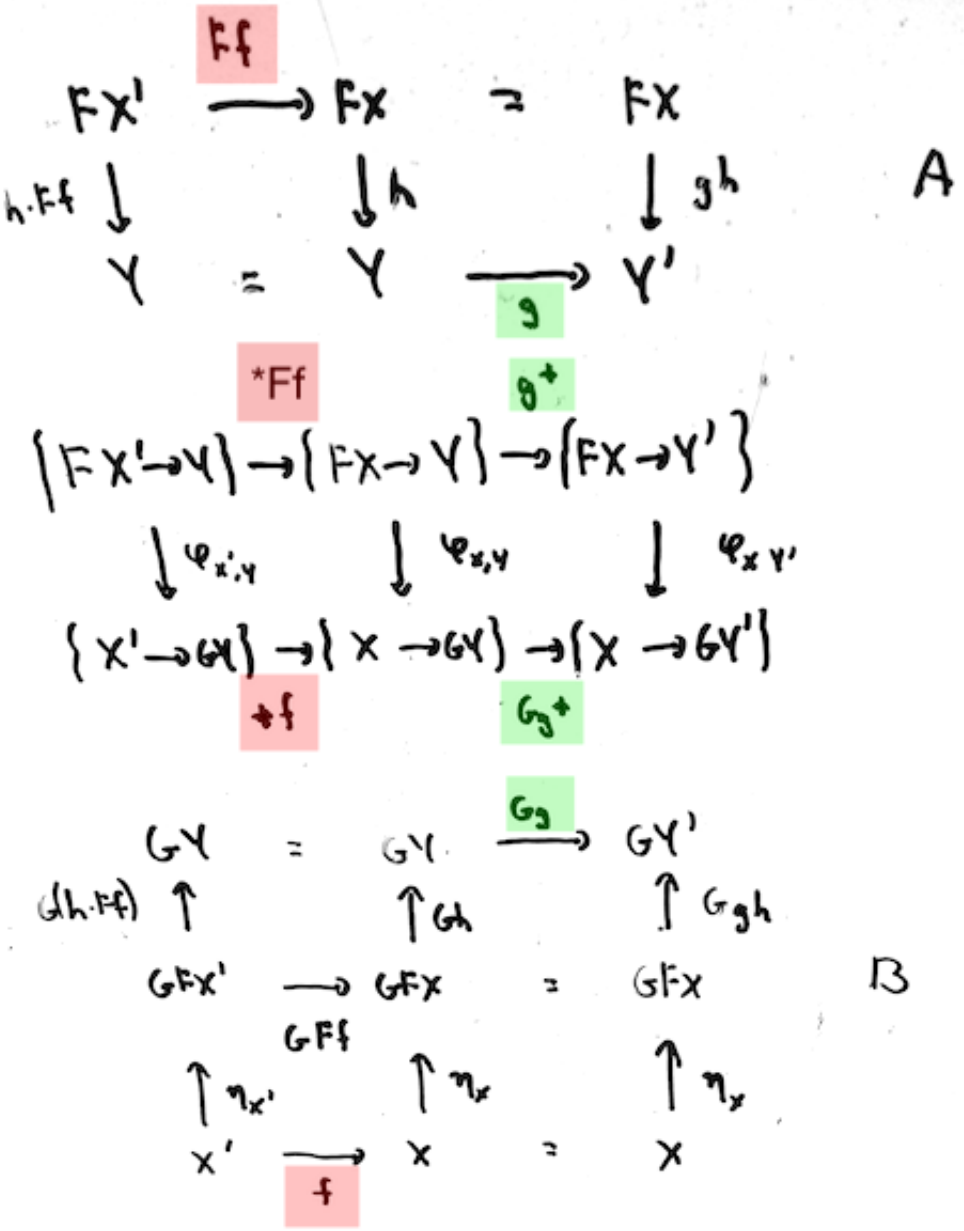
= h'



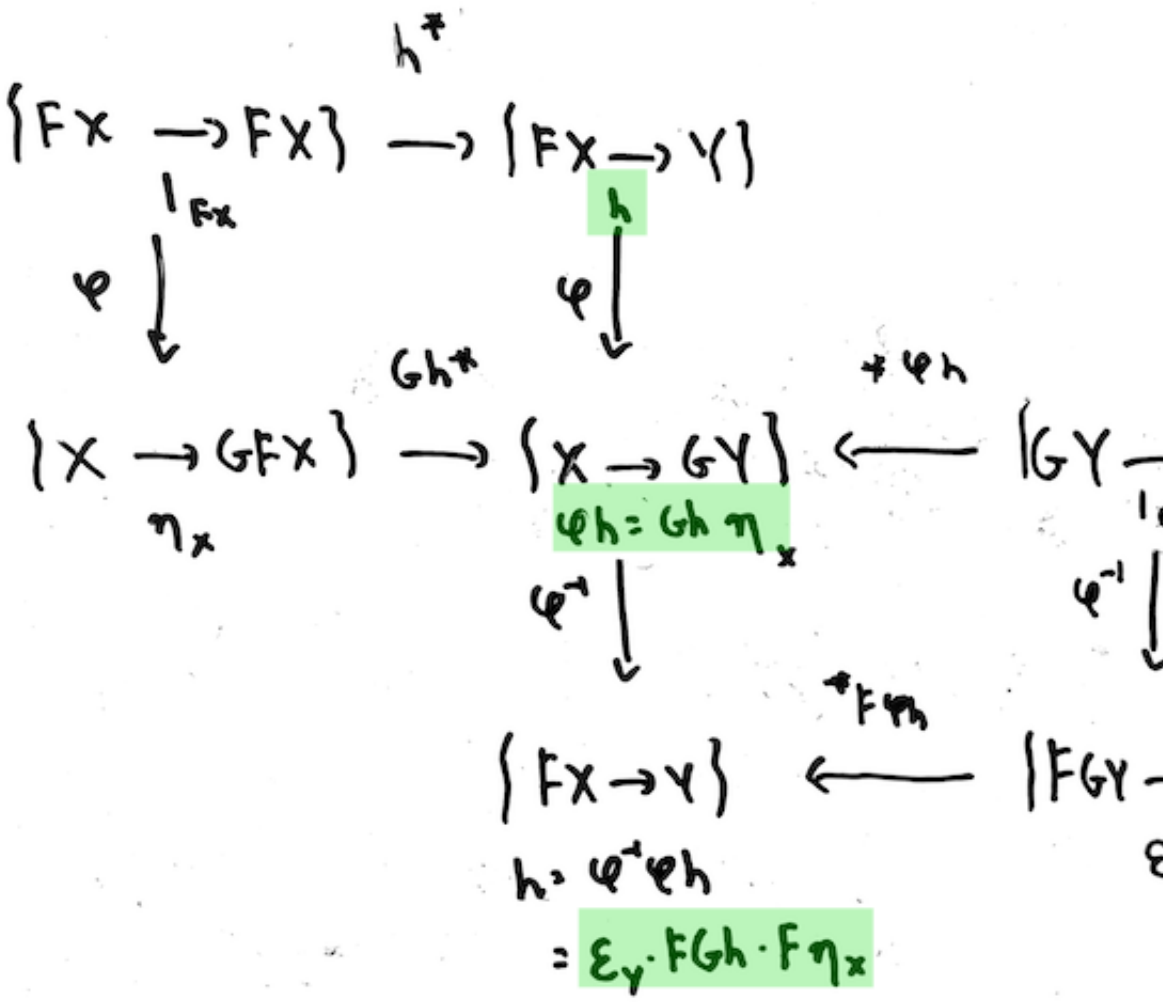
$-\varphi(\eta X)$	=	1GX	by definition
$G\varepsilon \circ \eta G$	=	IG: G ⇒ G	as natural transformation
$\varphi(\varepsilon Y)$	=	1FX	by definition
$\varepsilon F \circ F\eta$	=	IF: F ⇒ F	as natural transformation

- **naturality**

arrow form	in B	$\varphi \circ g^*$	=	$Gg^* \circ \varphi$
	in A	$\varphi \circ {}^*Ff$	=	${}^*f \circ \varphi$
function form	in B	$\varphi(g \circ h)$	=	$Gg \circ \varphi h$
	in A	$\varphi(h \circ Ff)$	=	$\varphi h \circ f$
universal form	in B	$G(g \circ h) \circ \eta X$	=	$Gg \circ (Gh \circ \eta X)$
	in A	$G(h \circ Ff) \circ \eta X$	=	$(Gh \circ \eta X') \circ f$
unit form	in B	$\varepsilon Y \circ g$	=	$FGg \circ \varepsilon Y'$
	in A	$GFf \circ \eta X$	=	$\eta X' \circ f$



Naturality + universality



Corollary: isomorphic functors
If <F, G, φ> and <F', G, φ'> are adjoints, then F and F' are isomorphic

==========
Proof
Universal properties

ηX: X→GFX	universal element of X to GF
η'X: X→GF'X	universal element of X to GF'

=> FX and F'X satisfy the same universal property

=> FX ≅ F'X with isomorphism

φ^-1(η'X): FX→FX' φ'^-1(ηX): FX'→FX

Let

ψ: F ⇒ F'
ψX: FX→FX'
ψX = φ^-1(η'X)

claim: ψ is an isomorphism
i.e. natural + each ψX is an isomorphism

Proof claim:
- isomorphism: by definition

- natural:
for any g: X→Y

ψY ◦ Fg	=	φ^-1(η'Y) ◦ Fg	
	=	φ^-1(η'Y ◦ g)	by composition
	=	φ^-1(GF'g ◦ η'X)	by naturality
	=	F'g ◦ φ^-1(η'X)	
	=	F'g ◦ ψX	

QED
==========

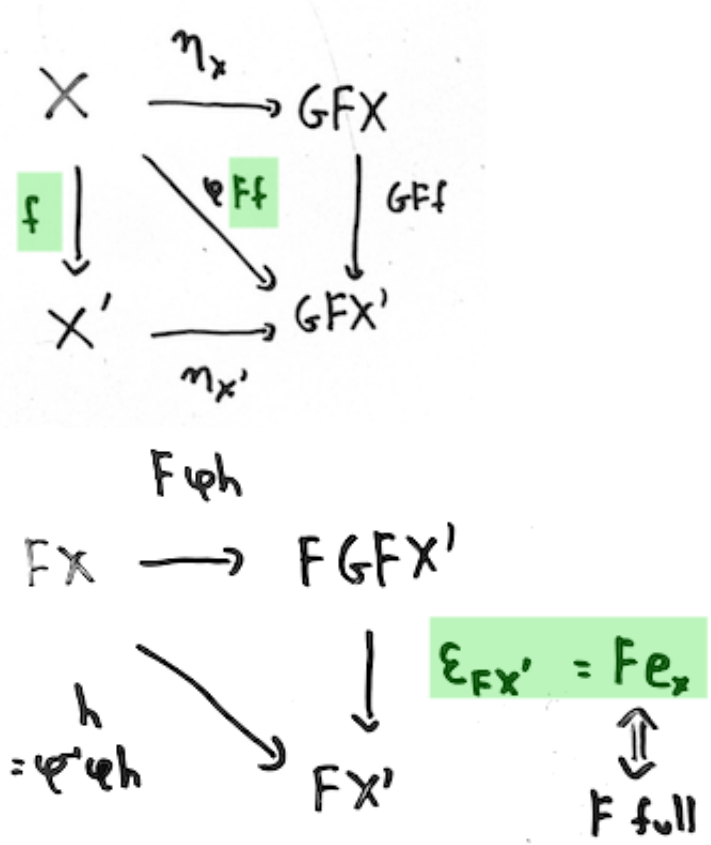
Proposition: preservation of isomorphisms
Let <F, G, φ>

φX.Y **preserves isomorphisms** <=> **unit ηX: X → GFX isomorphic**
i.e. h: FX → Y isomorphism
=> φh isomorphism

-φX.Y **preserves isomorphisms** <=> **counit** εY: FGY → Y **isomorphic**

Proof: universal property preserves isomorphisms

Full/faithful
Adjoints are a form of isomorphism between categories.
Hence it provides a way to define **full** and **faithful** in terms of **natural transformations**.



Proposition: full/faithful

F is **faithful** <=> η is **monic**

F is **full** <=> ε**FX = FeX** for any X
<=> η is **split epic**
$\eta X \circ eX = 1GFX$

G is **faithful** <=> ε is **epic**

G is **full** <=> η**GY = GnY** for any Y
<=> ε is **split monic**
$\varepsilon Y \circ nY = 1FGY$

Hence

F is **full** and **faithful** <=> η is a natural **isomorphism**

G is **full** and **faithful** <=> ε is a natural **isomorphism**

==========
Proof:

- **faithful +** η

$\eta X \circ f = \eta X' \circ GFf$ by naturality
$= \varphi Ff$

Hence

η is monic <=> $\varphi Ff = \varphi Ff'$ implies f =f '
<=> Ff = Ff' implies f = f' by 1-1 of adjoint
i.e. faithful

- **full** + η

Let h: FX → FX'. Use the adjoint to **map** it into F:

h = $-\varphi\varphi h$
$= \varepsilon FX' \circ F\varphi h$
$= F(eFX' \circ \varphi h)$ if $\varepsilon FX' = FeX$ for some eX': GFX' → X'

=>

F is **full** <=> $\varepsilon FX = FeX$ for some eX: GFX → X
<=> $1GFX = \eta$**X** $\circ$ **eX** by **naturality** $\eta X \circ eX$
$= G\varepsilon FX \circ \eta GFX = 1GFX$
i.e. ηX is split epic

- isomorphism

By general property, **monic + split epic** implies isomorphic
i.e.

$\eta X = (\eta X \circ eX) \circ \eta X$
$= \eta X \circ (eX \circ \eta X)$
$\eta X = \eta X \circ 1$
=> $(eX \circ \eta X) = 1$ since ηX is monic

QED
==========

e.g. **forgetful functor U**: faithful

Vect Set
{ Free X → V } <=> { X → UV }

U: Vect → Set	<=>	εV: Free(UV) → V
forgetful is faithful		**interpretation** is surjective
Free: Set → Vect	<=>	ηX: X → UFree(X)
free construction is faithful		**free injection** is injective

e.g. **injection functor I**: full + faithful

Ab **Grp**
{ G/[G, G] → A } <=> { G → IA } adjoint

I: Ab → Grp	<=>	εA: A/ [A, A] → A
injection is full + faithful		εA(a) = a
		injective homomorphism is isomorphic
Abelian: Grp → Ab	<=>	ηX: G → I(G/[G, G])
Abelian(G) = G/[G, G]		ηX(g) = g/[G, G]
Abelianization is full		Abelianization map is surjective

e.g. **Δ**X: A → AxA

A x A A
{ **Δ**X → (Y, Y') } <=> { X → YxY' }

Δ is faithful	<=>	ηX: X → XxX
		monic

==========

--- Binaturality ---

Binaturality
Adjoint is a **universal property** that is **binatural**
i.e. **natural in both** categories A and B

Theorem: naturality
Let bijection φX,Y: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is a **universal property** of FX <=> φ is **natural** in Y (as homsets)
φh = Gh ◦ φ1 φ(h◦1) ∈ φX,Y ◦ h* φX,Y' ◦ g* = Gg* ◦ φX,Y
Gh ◦ φ1 ∈ Gh* ◦ φX,FX

φ is an **adjoint** <=> φ is **natural** in X and Y (as homsets)
definition φX,Y' ◦ g* = Gg* ◦ φX,Y
φX,Y◦ *Ff = *f ◦ φX',Y

	<=>	φ is a **universal property**
natural in X		
		φX h = Gh ◦ ηX
universal		
		φX' h ◦ f = φX (h ◦ Ff) natural in
X		
	<=>	**universal arrow** is natural
(Gh ◦ φ1FX) ◦ f = G(h ◦ Ff) ◦ φ1FX'		ηX ◦ f = GFf ◦ ηX'

==========
Proof

- **case**: natural in X and Y <=> universal natural in X)

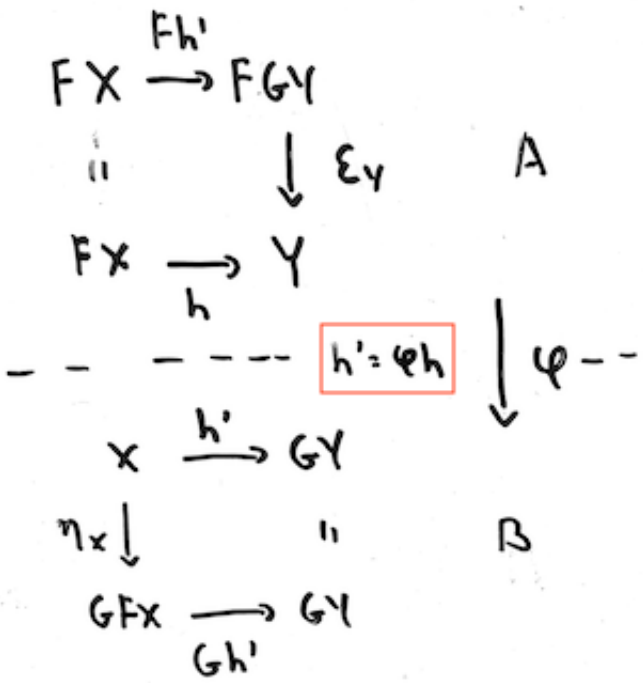
natural in Y
<=> universal property

natural in X
<=> *f ◦ φX,Y = φX',Y◦ *Ff
<=> φX,Yh ◦ f = φX',Y(h ◦ Ff)
i.e. natural in X

QED
==========

==========
--- Biuniversality ---

Biuniversality
Adjoint is **binuniversal**
i.e. universal in both directions φ and -φ.



Theorem: biuniversal
Let bijection φ: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is adjoint	<=>	φ is **biuniversal**	
		$\varphi g = Gg \circ \eta X$	universal
		$-\varphi f = \varepsilon Y \circ Ff$	couniversal

==========
Proof

natural in Y	<=>	universal
natural in X	<=>	couniversal

QED
==========

A universal property is determined completely by the **universal arrows** i.e. **units (Yoneda)**.

Theorem: biuniversal by units
Let bijection φ: {FX→Y} → {X→GY} with functors F: A → B, G: B → A

φ is **biuniversal** i.e. **adjoint** <=> **natural units** η: IA ⇒ GF, ε: FG ⇒ IB

$\varphi g = Gg \circ \eta X$	universal	$\varphi(\varepsilon Y) = G\varepsilon Y \circ \eta GY$	**IG = Gε ∘ ηG**
$-\varphi f = \varepsilon Y \circ Ff$	couniversal	$-\varphi(\eta X) = \varepsilon FX \circ F\eta X$	**IF = εF ∘ Fη**

==========
Proof
- adjoint => natural units

1GY	=	$\varphi(-\varphi(1GY))$
	=	$G(-\varphi(1GY)) \circ \eta GY$
	=	$G\varepsilon Y \cdot \eta GY$

=> IG = Gε ◦ ηG

and

φ is adjoint
=> universal arrow ηX is natural

Likewise with 1FX.

- natural units => adjoint
Idea: just combine with unit + use naturality to invert
For any h: X → GY

$\varphi(\varepsilon$Y ◦ Fh) =	G(εY ◦ Fh) ◦ ηX	**NOTE**: εY ◦ Fh = -φh	
	= GεY ◦ (GFh ◦ ηX)		
	= GεY ◦ ηGY ◦ h	natural	
	= h		

=> φ is **surjective**

Conversely, for any φg = Gg ◦ ηX

εY ◦ Fφg	= εY ◦ FGg ◦ FηX	NOTE: εY ◦ Fφg = -φ(φg)	
	= g ◦ εFX ◦ FηX	natural	
	= g		

=> φ is **injective**

=> φ is **bijective**, hence a universal property.

Likewise for -φ

QED
==========

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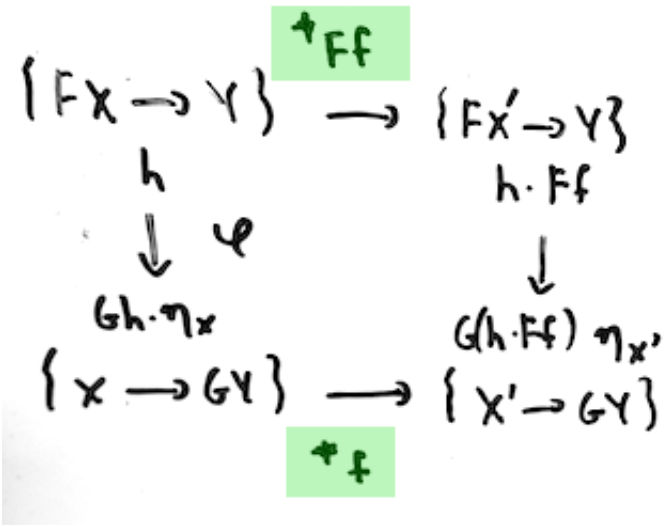
--- Characterization ---

Extension
A universal properties can be **extended** to be **binatural** i.e. natural in both directions

Proposition: binatural extension
For each X, let bijection φX,Y: {FX→Y} → {X→GY} with functor G: B → A

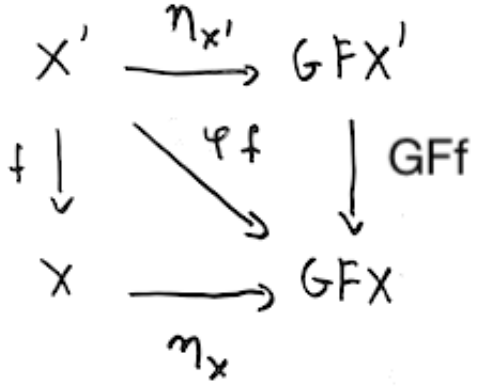
Then there is exactly one way to extend F to a **functor F: A → B** such that φ is also **natural in A.**
For any f: X' → X

φ is **natural in A**	<=>	**Ff: FX' → FX**
φh ◦ f = φ(h ◦ Ff)		**Ff = -φ(ηX ◦ f)**
(Gh ◦ ηX) ◦ f = G(h ◦ Ff) ◦ ηX'		



In such a case, Ff is precisely the natural arrow for the **universal arrows** η**X**:

η**X** ◦ **f** = **GFf** ◦ η**X'**



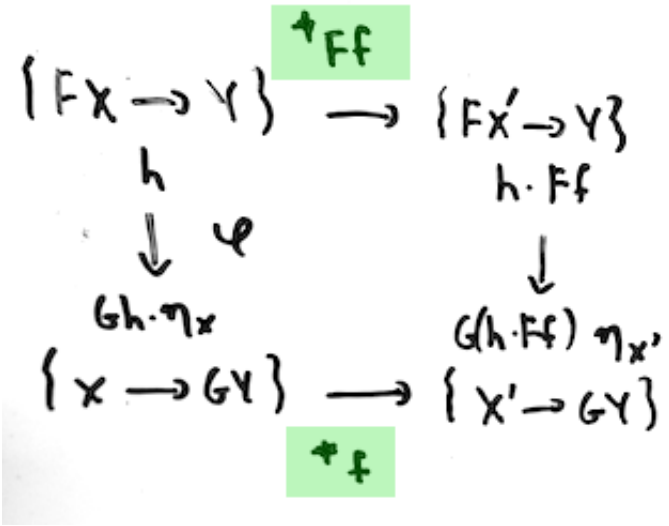
Corollary: extension
A universal property can be uniquely extended to be **binatural**
That is, an **adjoint**.

==========
Proof:

<=) Pick any h:FX → Y

φh ◦ f	=	(Gh ◦ ηX) ◦ f	
	=	Gh ◦ φFf	by assumption Ff = φ^-1(ηX ◦ F)
	=	φ(h ◦ Ff)	by naturality in B

=> φ is natural in A



=>) Suppose φ is natural in A.
Then

$\varphi(Ff \circ 1FX) = \varphi 1FX \circ f$
$= \eta X \circ f$

=> $Ff = \varphi^{-1}(\eta X \circ 1)$

- naturality of ηX

$\eta X \circ f = \varphi 1FX \circ f$
$= \varphi(1FX \circ Ff)$
$= GFf \circ \eta X'$

QED
==========

Characterization
Theorem: characterization

The following are equivalent

1) <F, G, φ>: A → B adjunction

2) universal property + **natural**:

F: A → B, G: B → A — functor
$\varphi X,Y$: {FX → Y} → {X → GY} — universal properties natural in both X and Y

3) universal property + **extension**:

G: B → A — functor
$\varphi X,Y$: {FX → Y} → {X → GY} — universal properties defined for each X

Adjoint if F is extended to a functor

functor F: A → B
Ff: FX→FX'
$Ff = -\varphi(\eta X \circ f)$ i.e. unique value such that
$f = GFf \circ \eta X$

4) universal property (via **universal element**) + **natural**:

F: A → B, G: B → A	functors
η**: IA ⇒ GF**	
ηX: X→GFX	universal element of {X→G_}

5) universal property (via **universal element**) + **extension**:

G: B → A	functor
FX	object for each X
η**X: X→GFX**	universal element of {X→G_}

Adjoint if F is extended to a functor

functor F: A → B	
Ff: FX→FX'	
Ff = f'	unique f' such that f = Gf' ηX

6) **natural unit + counit**:
Natural units determine a universal property

natural units
η**: IA ⇒ GF**
ε **: FG ⇒ IB**

Identity
εF ∘ Fη = IF
Gε ∘ ηG = IG

In all cases, the adjunction is defined from the unit η: IA ⇒ GF by

adjunction
φ: {FX→Y} → {X→GY}
φf = Gf ∘ ηX

Corollary (Counit)
Adjoints can be defined in terms of counits
i.e.
The following are equivalent

1) <F, G, φ> is an adjunction

2) via **counit**:

F, G	functors
ε: FG ⇒ IB	natural
εY: FGY → Y	couniversal element of {F_→Y}

3) via **unit**:

F	functor
G: B → A	object function

εY: FGY → Y natural
g ∘ εY = $\varepsilon Y'$ ∘ FGg i.e. natural in Y ∈ obj(B)

This induces a functor G

functor G
G: B → A
Gg: GY'→GY
Gg = g' unique g' such that g = εY ∘ Fg'

In all cases, the inverse adjunction is defined from the counit εY: FGY → Y by

adjunction
φ^-1: {X→GY} → {FX→Y}
φ^-1 = εY ∘ Fg

Limit
Adjuncts preserve **limits** by **1-1 correspondence** of arrows.

Theorem: limit preservation
Adjusts preserve limits
i.e. Let <F, G, φ>: A → B be an adjunction
For any shape T: J → A, S: J → B

Colim FT = F (Colim T) if Colim T exists
universal **Fu: FT => ΔF(Colim T)** universal u: T ⇒ **Δ**Colim T

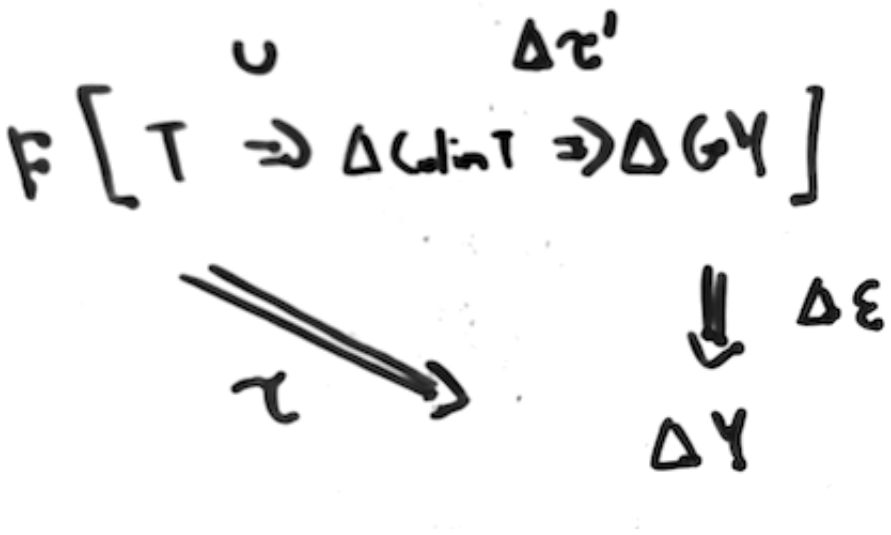
Lim GS = G (Lim S) if Lim S exists
universal **Gv: Δ G (Lim S) ⇒ GS** universal v: **Δ**Lim S ⇒ S

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Proof
Idea: adjoint preserves **cones**

Let τ: FT ⇒ **Δ**Y be any cone

$\varphi\tau$: T ⇒ **Δ**GY
$\varphi\tau$j = φ(τj)
=> unique τ': Colim T → GY
$\varphi\tau$ = **Δ**τ' ∘ u by colim

=>
τ = -φ ($\varphi\tau$)
= **Δ**εY ∘ F$\varphi\tau$
= **Δ**εY ∘ F**Δ**τ' ∘ Fu
= **Δ(**ε**Y** ∘ **F**τ**')** ∘ **Fu** since F**Δ**X = **Δ**FX for any X



- **uniqueness**: $\varepsilon Y \circ F\tau'$: F Colim T → Y is unique
If there is another ρ: F Colim T → Y such that

$\tau = \Delta\rho \circ Fu$

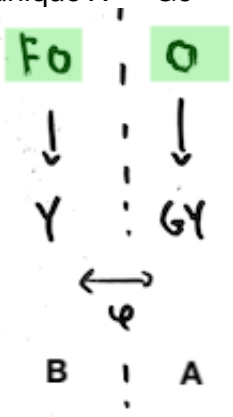
$\Rightarrow \tau = \Delta\rho \circ Fu$
$= \Delta(\varepsilon Y \circ F\varphi\rho) \circ Fu$ by adjoint $\rho = -\varphi(\varphi\rho) = \varepsilon Y \circ F\varphi\rho$
$= \Delta\varepsilon Y \circ F(\varphi\rho \circ u)$

QED
==========

Corollary

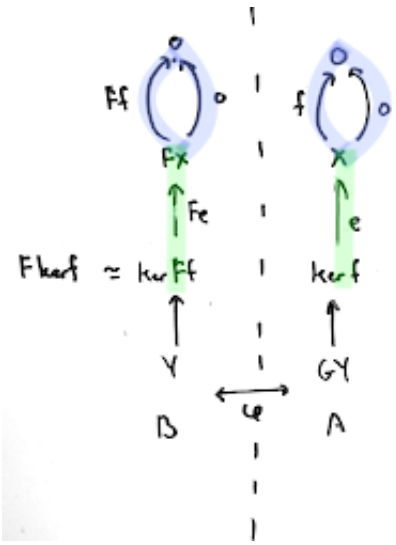
initial/terminal
F0 is **initial** if 0 is initial
unique F0 → Y fo

G0 is **terminal** if 0 is terminal
unique X → G0



monic/epic
Ff: FX → FX' is **epic** if f: X → X' is epic
Gg: GY → GY' is **monic** if g: Y → Y' is monic

kernel
ker Ff ≅ F(ker f)
Coker Gg ≅ G(Coker g)



product/coproduct

FX x FX' ≅ F (X x X') if X x X' ∈ A exists

projections projections (p,q): X x X' → (X, X')

Fp: FX x FX → FX

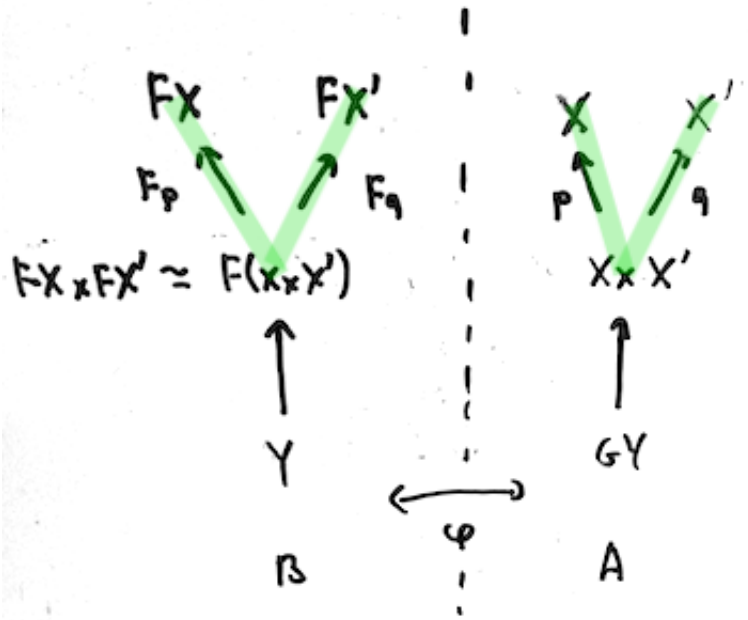
Fq: FX x FX' → FX'

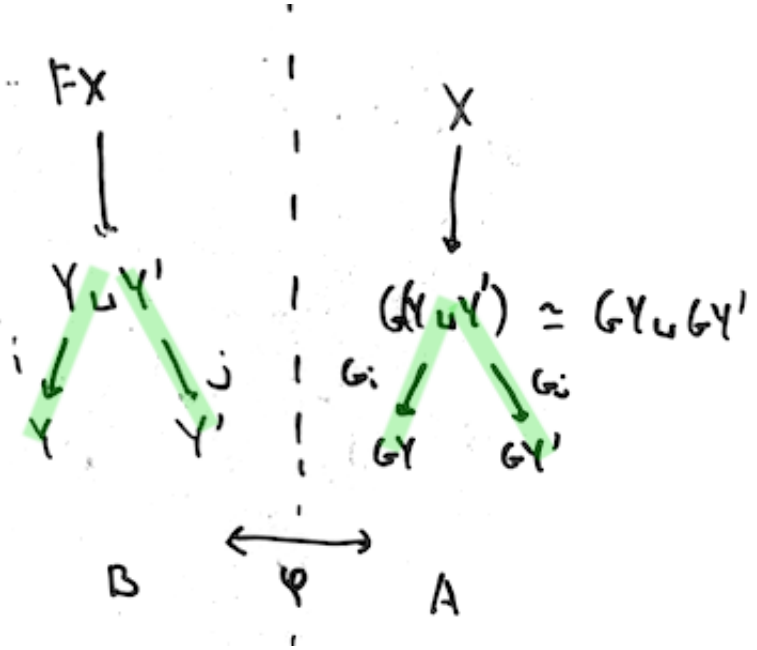
GY ⊔ GY' ≅ G (Y ⊔ Y') if Y ⊔ Y' ∈ B exists

injections injections (i, j): (Y, Y') → Y ⊔ Y'

Gi: GY → GY ⊔ GY'

Gj: GY → GY ⊔ GY'





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--- Equivalence ---

Equivalence is where F: A → B is an **isomorphism** with a **subcategory FA** ⊆ B.

Theorem: characterization
Let F: A → B

Equivalent statements:
1) F is an **equivalence** of categories

2) <F, G, ε, η> are **equivalent adjoints**

3) F is **full** and **faithful**
Each Y ∈ B is **isomorphic** to at least one object of FA.
i.e. can define GY ∈ FA such that **Y ≅ FGY**

Corollary: reflection
Let A' ⊆ A be a **full subcategory** of A.
If each X ∈ A is isomorphic to some FX ∈ A' with FX' = X' then the **reflection F** is an **equivalence**.
i.e.

<F, K, 1: IA' → IA', η: IA → KF>

==========
Proof:

2 => 3)
- full and faithful:
unit ηX: X → GFX is an isomorphism
=> F is faithful and full

- Y ∈ B
εY: FGY → Y is an isomorphism
=> Y ≅ FGY ∈ FA

3 => 2)
Idea: just construct a 1-1 correspondence with subcategory FA

Let Y ∈ B
Define GY ∈ FA by

εY: FGY → Y isomorphism

claim: εY is a unit of { X → GY } → { FX → Y }

proof claim:
Let h: FX → Y

=> εY^-1 ◦ h: FX → FGY

=> Fh' = εY^-1 ◦ h for some h: X → GY since F is full
h' is unique since F is faithful

=> 1-1 relationship h: FX → Y <=> h': X → GY
QED

Then

<F, G> an adjoint by extending G to a functor
εY: FGY → Y an **isomorphism** by definition
ηX: X → GFX an **isomorphism** inverse defined by
εFX ◦ FηX = 1FX
=> ηX = F^-1(εFX^-1) F^-1 exists

since F is faithful

ηX^-1 = F^-1(εFX)

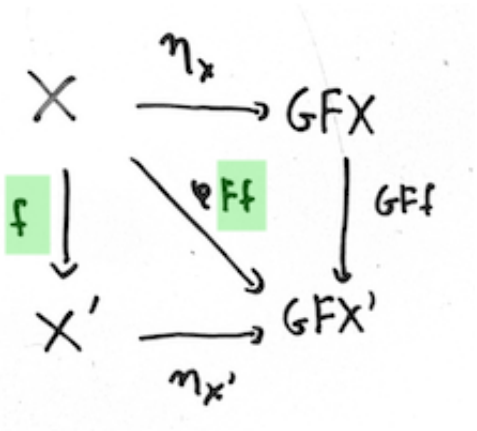
QED
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----- OLD -----

- **faithful**:
Let Ff: FX → FX'. **Invert** it to f with the adjoint:

φFf = GFf ◦ ηX
= ηX' ◦ f by naturality
=>
F is **faithful** <=> ηX is **monic**



- **full**:
Let h: FX → FX'. Use the adjoint to **map** it into F:

h	=	-φφh	
	=	εFX' ◦ Fφh	
	=	F(eFX' ◦ φh)	if εFX' = FeX for some eX': GFX' → X'

=>

F is **full**	<=>	εFX = FeX for some eX: GFX → X	
	<=>	1GFX = η**X** ◦ **eX**	by **naturality** ηX ◦ eX = GεFX ◦ ηGFX = 1GFX
		i.e. ηX is split epic	

Hence

Adjoint
Idea
A canonical notion of an **isomorphic functor F: A → B**
i.e. a **1-1 correspondence of morphisms** between A and B.

If F: A → B is bijective, then there is a 1-1 correspondence:

B		**A**
Fh: FX → FX'	<=>	h: X → X'

or for symmetry with G = F^-1

B		**A**
Fh: FX → Y	<=>	h: X → GY
Fh = εY ◦ Fh		h = G(Fh) ◦ ηX
unit 1Y = εY: FGY → Y		**unit** 1X = ηX: X → GFX

An **adjoint** is a generalization of this to any G: B → A and offsets with a 1-1 correspondence:

units
η: IA ⇒ GF ε: FG ⇒ IA

B		**A**
h': FX → Y	<=>	h: X → GY
h' = εY ◦ Fh		h = Gh' ◦ ηX

-

	Functor G: B → A	**Adjoint φ: B → A**
Arrows	G: { Y → Y' } → { GY → GY' } Gh: GY → GY'	φ: { FX → Y } → { X → GY } φh: X → GY
Composition	G(h ∘ f) = Gh ∘ Gf: GX → GY G(g ∘ h)= Gh ∘ Gh: GX → GZ where f: X → Y g: Y' → Z	φ(h ∘ Ff) = φh ∘ f: X' → GY φ(g ∘ h) = Gg ∘ φh: X → GY' where f: X' → X g: Y → Y'